

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 2

Year 12 Higher School Certificate
Trial Examination Term 3 2024

STUDENT NUMBER: _____ STUDENT NAME: _____ TEACHER: _____

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided separately
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination room

Total marks – 100

Section I Pages 3 – 7

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet provided

Section II Pages 8 – 14

90 marks

Attempt Questions 11 – 16

Start each question in a new writing booklet

Write your student number on every writing booklet

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/15	/15	/15	/15	/15	/15	/100

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment
Outcomes assessed: MEX12-1, MEX12-2, MEX12-3, MEX12-4, MEX12-7, MEX12-8, MEX12-9.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section.

Use the Objective Response answer sheet for Questions 1 – 10

1 Which of the following is the correct expression for $\int \frac{-1}{\sqrt{1-16x^2}} dx$?

- (A) $\frac{1}{4} \cos^{-1} \frac{x}{4} + C$
- (B) $4 \cos^{-1} \frac{x}{4} + C$
- (C) $\frac{1}{4} \cos^{-1} 4x + C$
- (D) $4 \cos^{-1} 4x + C$

2. If $\underline{a}$, $\underline{b}$ and $\underline{c}$ are the position vectors of the three vertices of a triangle, and $(\underline{b} - \underline{a}) = \lambda(\underline{c} - \underline{a})$ where $0 < \lambda < 1$ which of the following must be false?

- (A) Point B is dividing the interval AC in the ratio of $\lambda : (1 - \lambda)$ internally.
- (B) Point B is dividing the interval AC in the ratio of $\lambda : (1 - \lambda)$ externally.
- (C) Points A , B and C are collinear points.
- (D) $\overrightarrow{AB} \parallel \overrightarrow{AC}$.

3 Which of the following statements is true?

- (A) $\forall a, b \in \mathbb{R} \quad \sin a = \sin b \Rightarrow a = b$
- (B) $\forall a, b \in \mathbb{R} \quad |a + b| > |a - b|$
- (C) $\exists a, b \in \mathbb{R} \quad \text{such that } \log_e(a + b) = \log_e(ab)$
- (D) $\exists a, b \in \mathbb{C} \quad |a + b| > |a| + |b|$

4. Which of the following is the correct expression for $\int x \tan^{-1} x \, dx$

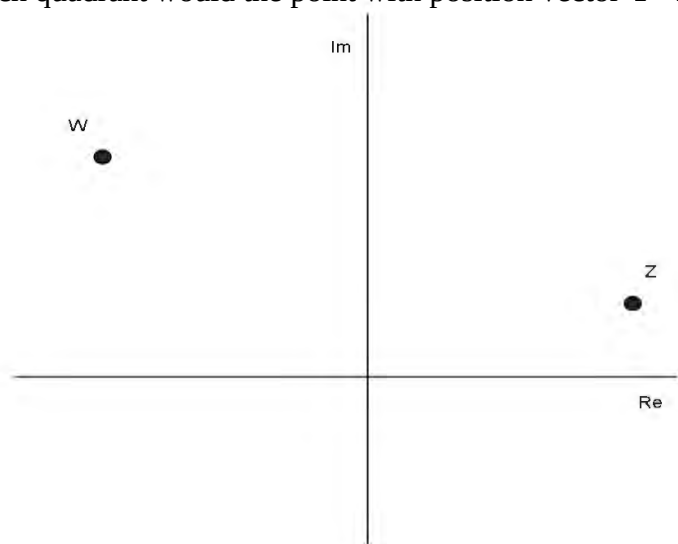
(A) $x + \tan^{-1} x + C$

(B) $\frac{x}{\tan^{-1} x} + \frac{1}{2} \ln(1 + x^2) + C$

(C) $x \tan^{-1} x + \frac{1}{2} \ln(1 + x^2) + C$

(D) $\frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C.$

5 The diagram below shows the position of points z and w , that have position vectors z and w respectively. In which quadrant would the point with position vector $\vec{z} - iw$ lie?



(A) Quadrant 1

(B) Quadrant 2

(C) Quadrant 3

(D) Quadrant 4

6. Consider the statement:

‘For any function $f(x)$, $f(x)$ is not continuous at $x = a \Rightarrow f(x)$ is not differentiable at $x = a$.’

Which of the following statement is correct?

- (A) The converse statement is false and the contrapositive statement is false.
- (B) The converse statement is false and the contrapositive statement is true.
- (C) The converse statement is true and the contrapositive statement is false.
- (D) The converse statement is true and the contrapositive statement is true.

7 Consider the vectors $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\overrightarrow{OB}$ with $|\overrightarrow{OB}| = 3$.

Given that $\overrightarrow{OA} \cdot \overrightarrow{OB} = 6$, find in square units, the area of ΔOAB .

(A) $3\sqrt{10}$

(B) $\frac{3\sqrt{14}}{2}$

(C) $\frac{3\sqrt{10}}{4}$

(D) $\frac{3\sqrt{10}}{2}$

8. If $\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = \frac{\pi}{2}$, what is the value of $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$?

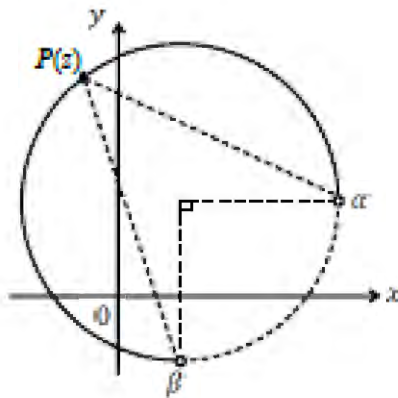
(A) 0

(B) $\frac{\pi}{2}$

(C) $\frac{\pi^2}{2}$

(D) $\frac{\pi^2}{4}$

- 9 The diagram shows the solution of an equation as traced out by the point $P(z)$. The path traced out by the point P representing the complex number z is three-quarter circle.



Which of the following could be the equation?

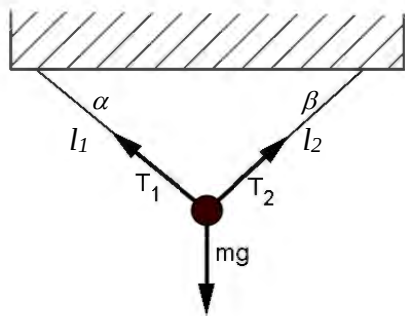
(A) $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = 0$

(B) $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = \frac{\pi}{4}$

(C) $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = \frac{\pi}{2}$

(D) $\text{Arg}(z - \beta) - \text{Arg}(z - \alpha) = \frac{\pi}{4}$

10. The mass m kg is suspended from the ceiling by two light strings of equal length, where the tensions in each string are equal.



If the lengths of the strings are changed such that $l_1 > l_2$, and the angles made by the strings to the horizontal are α and β respectively, which of the following will now be true?

- (A) $\alpha > \beta$, $T_1 > T_2$
- (B) $\alpha > \beta$, $T_2 > T_1$
- (C) $\alpha < \beta$, $T_2 > T_1$
- (D) $\alpha < \beta$, $T_1 > T_2$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section.

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet.

- (a) Let $w = 8 - 2i$ and $z = -5 + 3i$. Find $w + \bar{z}$ 1
- (b) (i) Show that $(1 - 2i)^2 = -3 - 4i$ 1
- (ii) Hence solve the equation $z^2 - 5z + (7 + i) = 0$ 3
- (c) (i) Find $\frac{d}{dx}(x \sin^{-1} x)$ 1
- (ii) Hence or otherwise find $\int \sin^{-1}(x) dx$ 2
- (d) Provide a non-inductive proof as to why $12^n > 5^n + 7^n$, $\forall$ integers $n \geq 2$. 2
- (e) (i) Use the result $e^{in\theta} = \cos n\theta + i \sin n\theta$ to show $e^{ni\theta} + e^{-ni\theta} = 2 \cos n\theta$ 1
- (ii) By expanding $(e^{i\theta} + e^{-i\theta})^4$, prove that $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$ 2
- (iii) Hence find $\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$ 2

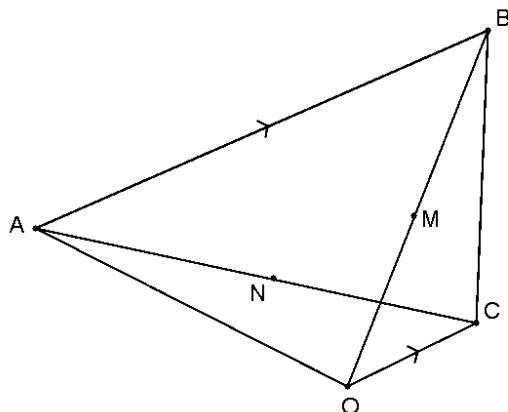
End of Question 11.

QUESTION 12 (15 marks)

(a) $OABC$ is a trapezium with $\overrightarrow{AB} = k\overrightarrow{OC}$.

Let $\underline{a} = \overrightarrow{OA}$ and $\underline{c} = \overrightarrow{OC}$.

M and N are midpoints of OB and AC respectively.



(i) Find $\overrightarrow{OM}$ in terms of $\underline{a}$ and $\underline{c}$. 2

(ii) Find $\overrightarrow{ON}$ in terms of $\underline{a}$ and $\underline{c}$. 2

(iii) Hence find $\overrightarrow{MN}$ in terms of $\underline{a}$ and $\underline{c}$. 1

(iv) Deduce the value of k required for $ABMN$ to be a parallelogram. 2

(b) (i) Express $\frac{x^2+1}{(x-1)(x^2+x+1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$. 2

(ii) Hence find $\int \frac{x^2+1}{(x-1)(x^2+x+1)} dx$ 3

(c) Prove using contradiction that $\log_2 5$ is an irrational number. 3

End of Question 12

Question 13 (15 marks) Start a new writing booklet.

- (a) Consider a sphere S , with centre $C(2, -1, 0)$ and radius $\sqrt{29}$ units. Consider also the line l with parametric equations

$$x = \lambda + 1, y = \lambda, z = 2\lambda + 3$$

- (i) Find the vector equation of line l , writing your answer in the form $\vec{r} = \vec{a} + \lambda\vec{b}$, where $\vec{a}$ and $\vec{b}$ are expressed as column vectors.

It is also known that l intersects the surface of S at points P and Q

- (ii) Find the coordinates of points P and Q .

- (iii) Hence or otherwise, determine whether PQ is a diameter of S , showing necessary working.

- (b) The displacement, x metres, of a particle P from the origin O at time t seconds is given by

$$x = 6 \cos\left(2t + \frac{\pi}{4}\right) + \cos(2t)$$

- (i) Show that P is moving in simple harmonic motion about O

- (ii) Find the amplitude of this motion, correct to 1 decimal place.

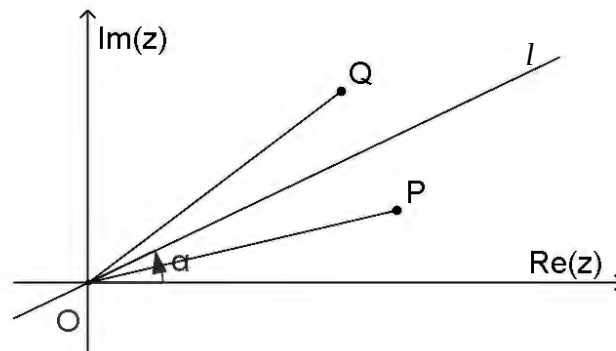
- (c) Using Trigonometric substitution, or otherwise, find $\int \frac{\sqrt{1-x^2}}{1-x^2} dx$.

Give your answer without trigonometric functions.

End of Question 13

QUESTION 14 (15 marks)

- (a) Let l be the line in the complex plane that passes through the origin and makes an angle α with the positive real axis, where $0 < \alpha < \frac{\pi}{2}$.



The point P represents the complex number z_1 , where $0 < \arg(z_1) < \alpha$. The point P is reflected in the line l to produce the point Q , which represents the complex number z_2 . Hence $|z_1| = |z_2|$.

- (i) Explain why $\arg(z_1) + \arg(z_2) = 2\alpha$. 2
 - (ii) Deduce that $z_1 z_2 = |z_1|^2 (\cos 2\alpha + i \sin 2\alpha)$. 1
 - (iii) Let $\alpha = \frac{\pi}{4}$ and let R be the point that represents the complex number $z_1 z_2$. 1
Describe the locus of R as z_1 varies.
- (b) If a , b and c are real and unequal and that $a^2 + b^2 > 2ab$ deduce that
- (i) $a^2 + b^2 + c^2 > ab + bc + ac$. 1
 - (ii) If $a + b + c = 6$ show that $ab + bc + ac < 12$. 2

Question 14 continues over the page

(c) Use the substitution of $t = \tan \frac{x}{2}$, find $\int \frac{1}{3 - \cos x - 2 \sin x} dx$ **3**

(d) A particle, initially at $x = 2$, has a velocity given by $v = \sqrt{16 - 3x^2}$. Find the expression of x in terms of t . **3**

(e) If the displacement of an object at any time t is given by the vector equation **2**

$$\vec{r} = \begin{bmatrix} 10 \cos t \\ 10 \sin t \\ 15 - t \end{bmatrix}$$

Find the vector equation of the velocity and its initial speed.

End of Question 14

Question 15 (15 marks) Start a new writing booklet.

(a) Consider the equation $z^5 + 1 = 0$, where z is a complex number.

(i) Solve the equation $z^5 + 1 = 0$ by finding the 5th roots of -1 . **2**

(ii) Show that if z is a solution of $z^5 + 1 = 0$ and $z \neq 1$, then $u = z + \frac{1}{z}$ **2**
is a solution of $u^2 - u - 1 = 0$.

(iii) Hence find the exact value of $\cos \frac{3\pi}{5}$. **3**

(b) Consider the polynomial equation

$$x^4 + ax^3 + bx^2 + cx + d = 0$$

Where a, b, c and d are all integers.

Suppose also that the equation has a root of the form ki , where k is real, and $k \neq 0$.

(i) State why the conjugate, $-ki$ is also a root. **1**

(ii) Show that $c = k^2a$. **2**

(iii) Show that $c^2 + a^2d = abc$. **2**

(iv) If 2 is also a root of the equation, and $b = 0$, show that c is even. **3**

End of Question 15

QUESTION 16 (15 marks)

- (a) Consider the recurrence relation defined by $T_1 = 1$ and

$$T_{n+1} = \frac{4+T_n}{1+T_n}, \text{ for } n = 1, 2, 3, \dots$$

- (i) Prove by mathematical induction that for $n \geq 1$,

4

$$T_n = 2 \left[\frac{1+(-3)^{-n}}{1-(-3)^{-n}} \right].$$

- (ii) Hence find the limit value of T_n as $n \rightarrow \infty$.

1

- (b) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ and let $J_n = (-1)^n I_{2n}$

- (i) Show that $I_n + I_{n+2} = \frac{1}{n+1}$.

2

- (ii) Find the value of $I_4 = \int_0^{\frac{\pi}{4}} \tan^4 x \, dx$

2

- (iii) Deduce that $J_n - J_{n-1} = \frac{(-1)^n}{2n-1}$ for $n \geq 1$.

1

- (iii) By considering

2

$$J_m = (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3}) + \dots + (J_1 - J_0) + J_0$$

Show that

$$J_m = \frac{\pi}{4} + \sum_{n=1}^m \frac{(-1)^n}{2n-1}, \text{ for } n \geq 1.$$

- (v) Use substitution $u = \tan x$ to show that $I_n = \int_0^1 \frac{u^n}{1+u^2} \, du$.

1

- (vi) Deduce that $0 \leq I_n \leq \frac{1}{n+1}$ and conclude that $J_n \rightarrow 0$ as $n \rightarrow \infty$.

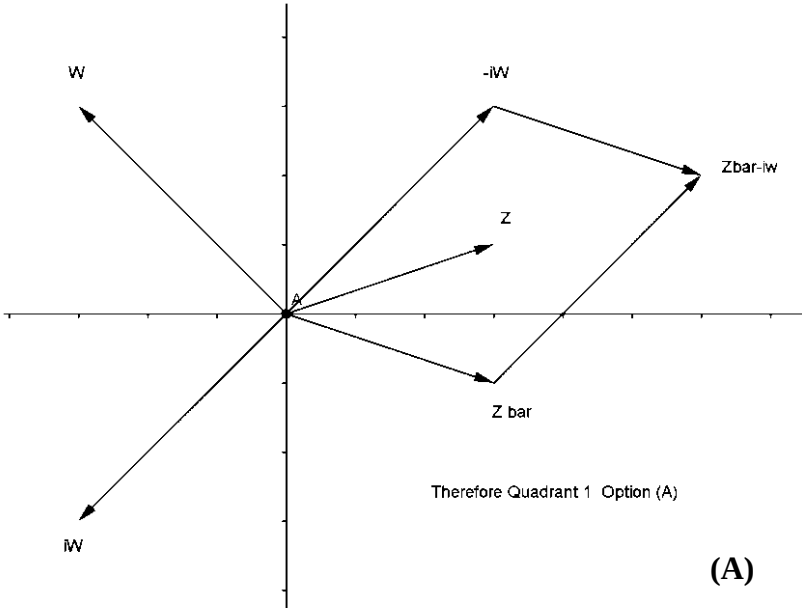
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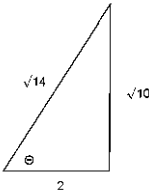
End of Examination

Section I Multiple Choice

Question	Answers
1	C
2	B
3	C
4	D
5	A
6	B
7	D
8	D
9	B
10	C

1.	$\int \frac{-1}{\sqrt{1-16x^2}} dx = \frac{1}{4} \int \frac{-1}{\sqrt{\frac{1}{16} - x^2}} dx$ $= \frac{1}{4} \cos^{-1} \frac{x}{\left(\frac{1}{4}\right)} + C$ $= \frac{1}{4} \cos^{-1} 4x + C \quad \text{(C)}$	90% correct.
2.	$(\underline{b}-\underline{a}) = \lambda(\underline{c}-\underline{a})$ has more information than $\overline{AB} \parallel \overline{AC}$.  Point B is dividing the interval AC in the ratio of $\lambda : (1-\lambda)$ internally. (B)	61% correct. Popular choice was A. Question required false statement. Upon reflection we will allow (c) as well as a correct answer
3.	(a) $\forall$ means "for all" $a = 30$ $b = 150$ not true (b) $\forall$ means "for all" $a = 5$ $b = -3$ not true (c) $\exists$ means "there exists" $a = 2$ $b = 2$ is true (d) $\exists$ means "there exists" $a+b > a + b$ not true as $a + b \geq a+b$ by Δ inequality $\forall \mathbb{C}$ (C)	74% correct. Popular choice was D. Third side of a triangle is less than the sum of the two sides.

4.	$\int x \tan^{-1} x \, dx = uv - \int u'v \, dx \quad \text{where } u = \tan^{-1} x \quad v' = x$ $u' = \frac{1}{1+x^2} \quad v = \frac{x^2}{2}$ $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx$ $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2 + 1 - 1}{1+x^2} \, dx$ $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2 + 1}{1+x^2} - \frac{1}{1+x^2} \, dx$ $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int 1 - \frac{1}{1+x^2} \, dx$ $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left(x - \tan^{-1} x \right) + C$ $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C \quad \text{(D)}$	100% correct.
5.	 Therefore Quadrant 1 Option (A) (A)	87% correct. Popular choice was D.

6.	Original statement: 'For any function $f(x)$, $f(x)$ is not continuous at $x = a \Rightarrow f(x)$ is not differentiable at $x = a$.' Converse: For any function $f(x)$, $f(x)$ is not differentiable at $x = a \Rightarrow f(x)$ is not continuous at $x = a$.' False. Contrapositive: For any function $f(x)$, $f(x)$ is differentiable at $x = a \Rightarrow f(x)$ is continuous at $x = a$.' True. (B)	68% correct. Popular choice was D.
7.	$\overline{OA} = \sqrt{4+1+9} = \sqrt{14}$ As $\overline{OA} \cdot \overline{OB} = \overline{OA} \cdot \overline{OB} \cos \theta$ then $6 = \sqrt{14} \times 3 \times \cos \theta$ $\therefore \cos \theta = \frac{2}{\sqrt{14}}$ Since $\overline{OA} \cdot \overline{OB} > 0$ and θ is in a triangle, then θ is acute. Using the triangle, $\sin \theta = \frac{\sqrt{10}}{\sqrt{14}}$ $\therefore \text{Area} = \frac{1}{2} \times 3 \times \sqrt{14} \times \frac{\sqrt{10}}{\sqrt{14}}$ $= \frac{3\sqrt{10}}{2}$  (D)	81% correct. Popular choice was B.

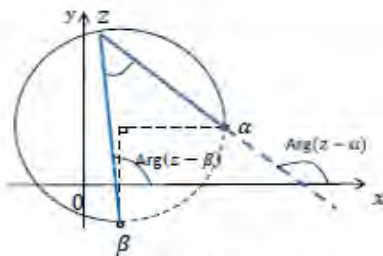
8.

$$\begin{aligned}
 I &= \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\
 &= \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx \\
 &= \int_0^{\pi} \frac{\pi \sin(\pi - x)}{1 + \cos^2(\pi - x)} - \frac{x \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx \\
 &= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} - \frac{x \sin x}{1 + \cos^2 x} dx \\
 &= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\
 I &= \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx - I \\
 2I &= \pi \left(\frac{\pi}{2} \right) \\
 2I &= \frac{\pi^2}{2} \\
 \therefore I &= \frac{\pi^2}{4}
 \end{aligned}$$

(D)

Poorly done.
48% correct.
Popular choice was C. Students
forgotten it was $2I$ that equals
 $\frac{\pi^2}{2}$.

9.



z lies on the circumference of a circle, radii from α and β meet at the centre at $\frac{\pi}{2}$ radians. This means the angle at z must be $\frac{\pi}{4}$ radians as the angle at the circumference is half the angle at the centre when subtended by the same arc.

OR

$z - \alpha$ represents the vector from α to z .

$z - \beta$ represents the vector from β to z .

Using the exterior angle of a triangle, we can see that the angle at $z = \text{Arg}(z - \alpha) - \text{Arg}(z - \beta)$

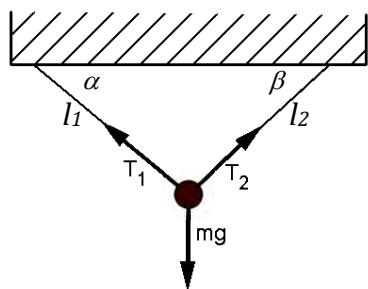
So, $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = \frac{\pi}{4}$

Hence, the correct option is B.

(B)

100% correct.

10.



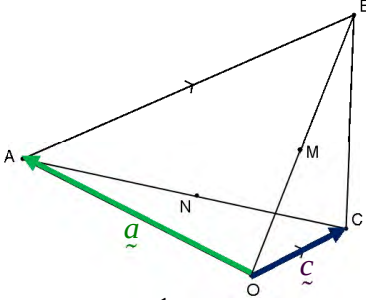
When $l_1 > l_2$, then $\beta > \alpha$ and the shorter string will experience greater tension, i.e. $T_2 > T_1$. (C)

Poorly done.
58% correct.
Popular choice was D.

SECTION 2

11.	(a)	$\bar{w} + \bar{z} = 8 - 2i + (-5 - 3i)$ ① $= 3 - 5i$	Well done
11.	(b)(i)	$(1 - 2i)^2 = 1 - 4i - 4$ ① $= -3 - 4i$	Well done
11.	(b)(ii)	① $z = \frac{5 \pm \sqrt{(-5)^2 - 4 \cdot 1 \cdot (7 + i)}}{2 \cdot 1}$ $= \frac{5 \pm \sqrt{25 - 4(7 + i)}}{2}$ ① $= \frac{5 \pm \sqrt{-3 + 4i}}{2}$ $= \frac{5 \pm \sqrt{(1 - 2i)^2}}{2}, \text{ from (i)}$ $= \frac{5 \pm (1 - 2i)}{2}$ $= 3 - i, 2 + i$ ①	Well done
11.	(c)(i)	$\frac{d}{dx}(x \sin^{-1} x) = x \cdot \frac{1}{\sqrt{1 - x^2}} + \sin^{-1} x \cdot 1$ $= \frac{x}{\sqrt{1 - x^2}} + \sin^{-1} x$ ①	Well done
11.	(c)(ii)	$\int \frac{d}{dx}(x \sin^{-1} x) dx = \int \frac{x}{\sqrt{1 - x^2}} dx + \int \sin^{-1}(x) dx$ $\therefore \int \sin^{-1}(x) dx = \int \frac{d}{dx}(x \sin^{-1} x) dx - \int \frac{x}{\sqrt{1 - x^2}} dx$ ① $= x \sin^{-1} x + \sqrt{1 - x^2} + C$ ①	Mostly well done

11.	(d) $d) 12^n = (5 + 7)^n$ $= \binom{n}{0} 5^n + \binom{n}{1} 5^{n-1} \times 7 + \binom{n}{2} 5^{n-2} \times 7^2 +$ $\dots + \binom{n}{n-1} \times 5 \times 7^{n-1} + \binom{n}{n} 7^n \quad \textcircled{1}$ As $\binom{n}{0} = \binom{n}{n} = 1$ then $(5 + 7)^n = 5^n + \binom{n}{1} 5^{n-1} \times 7 + \binom{n}{2} 5^{n-2} \times 7^2 +$ $\dots + \binom{n}{n-1} \times 5 \times 7^{n-1} + 7^n$ Since $\binom{n}{1} 5^{n-1} \times 7 + \binom{n}{2} 5^{n-2} \times 7^2 +$ $\dots + \binom{n}{n-1} \times 5 \times 7^{n-1} \text{ is a sum of positive terms} \quad \textcircled{1}$ then $(5 + 7)^n = 5^n + 7^n + \text{Sum of positive terms}$. This indicates that $(5 + 7)^n > 5^n + 7^n$; that is, $12^n > 5^n + 7^n$ for all integers $n \geq 2$.	Mostly well done
11.	(e)(i) $e^{in\theta} + e^{-in\theta} = [\cos(n\theta) + i \sin(n\theta)] + [\cos(-n\theta) + i \sin(-n\theta)]$ $= \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta) \quad \textcircled{1}$ $= 2 \cos(n\theta)$	well done
11.	(e)(ii) $(e^{i\theta} + e^{-i\theta})^4 = e^{i4\theta} + 4e^{i2\theta} + 6 + 4 + 4e^{-i2\theta} + e^{-i4\theta}$ $= (e^{i4\theta} + e^{-i4\theta}) + 4(e^{i2\theta} + e^{-i2\theta}) + 6$ $\therefore (2 \cos \theta)^4 = 2 \cos 4\theta + 4(2 \cos 2\theta) + 6 \quad \textcircled{1}$ $\therefore 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$ $\therefore \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} \quad \textcircled{1}$	well done

11.	(e)(iii) $\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = \int_0^{\frac{\pi}{2}} \left(\frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} \right) d\theta$ $= \left[\frac{1}{32} \sin 4\theta + \frac{1}{4} \sin 2\theta + \frac{3\theta}{8} \right]_0^{\frac{\pi}{2}}$ $= \left[\frac{1}{32} \sin 4\left(\frac{\pi}{2}\right) + \frac{1}{4} \sin 2\left(\frac{\pi}{2}\right) + \frac{3\left(\frac{\pi}{2}\right)}{8} \right] - \left[\frac{1}{32} \sin 4(0) + \frac{1}{4} \sin 2(0) + \frac{3(0)}{8} \right]$ $= \frac{3\pi}{16}$	Mostly well done ① ①
12.	(a)(i)  $\overrightarrow{OM} = \frac{1}{2} \overrightarrow{OB}$ $= \frac{1}{2} (\overrightarrow{OA} + \overrightarrow{AB}) \text{ but } \overrightarrow{AB} = k \overrightarrow{OC} \quad \text{① for } \overrightarrow{OB}$ $= \frac{1}{2} (\overrightarrow{OA} + k \overrightarrow{OC})$ $\therefore \overrightarrow{OM} = \frac{1}{2} \underline{a} + \frac{1}{2} k \underline{c} \quad \text{① for } \overrightarrow{OM}$	Mostly did well!
12.	(a)(ii) $\overrightarrow{ON} = \overrightarrow{OC} + \frac{1}{2} \overrightarrow{CA}$ $= \overrightarrow{OC} + \frac{1}{2} (\overrightarrow{OA} - \overrightarrow{OC}) \quad \text{① for } \overrightarrow{CA}$ $= \underline{c} + \left(\frac{1}{2} \underline{a} - \frac{1}{2} \underline{c} \right)$ $\therefore \overrightarrow{ON} = \frac{1}{2} \underline{a} + \frac{1}{2} \underline{c} \quad \text{① for } \overrightarrow{ON}$ OR $\overrightarrow{ON} = \frac{1}{2} (\overrightarrow{OA} + \overrightarrow{OC}) \quad \text{① for } \overrightarrow{CA}$ $\therefore \overrightarrow{ON} = \frac{1}{2} \underline{a} + \frac{1}{2} \underline{c} \quad \text{① for } \overrightarrow{ON}$	Mostly did well!
12.	(a)(iii) $\therefore \overrightarrow{MN} = \overrightarrow{ON} - \overrightarrow{OM}$ $= \left(\frac{1}{2} \underline{a} + \frac{1}{2} \underline{c} \right) - \left(\frac{1}{2} \underline{a} + \frac{1}{2} k \underline{c} \right)$	All did well!

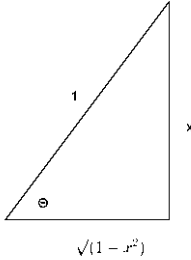
	$= \frac{1}{2}\underline{c} - \frac{1}{2}k\underline{c}$ $\therefore \overrightarrow{MN} = \frac{1}{2}\underline{c}(1-k)$ ① for $\overrightarrow{MN}$	
12.	(a)(iv) For $ABMN$ to be a parallelogram, $\overrightarrow{MN} = \overrightarrow{BA}$ ① $\frac{1}{2}\underline{c}(1-k) = -k\underline{c}$ $\frac{1}{2}(1-k) = -k$ $1-k = -2k$ $1-k = 2k$ $1 = -k$ $k = -1$ ① for k OR For $ABMN$ to be a parallelogram, such that $\overrightarrow{MN} = -\overrightarrow{AB}$ ① $\left \frac{1}{2}\underline{c}(1-k) \right = k \underline{c} $ $\frac{1}{2} (1-k) \underline{c} = k \underline{c} \quad \text{since } 1-k < 0, k > 1$ $\therefore (1-k) = k-1$ $\frac{1}{2}(k-1) = k$ $k-1 = 2k$ $-1 = k$ $\therefore k = -1$ ① for k	Not as well executed. Students who paid attention to direction of vectors get the correct answers.
12.	(b)(i) $\frac{x^2+1}{(x-1)(x^2+x+1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$ $x^2+1 \equiv A(x^2+x+1) + (Bx+C)(x-1)$ When $x=1$, $(1)^2+1 \equiv A((1)^2+(1)+1)+0$ $2=3A$ $A=\frac{2}{3}$ When $x=0$, $(0)^2+1 \equiv \frac{2}{3}((0)^2+(0)+1) + (B(0)+C)((0)-1)$ $1 = \frac{2}{3} - C$ $C = \frac{2}{3} - 1$ $C = -\frac{1}{3}$	Mostly did well! A few arithmetic calculation error. A few students did not answer as required in $\frac{x^2+1}{(x-1)(x^2+x+1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$ form.

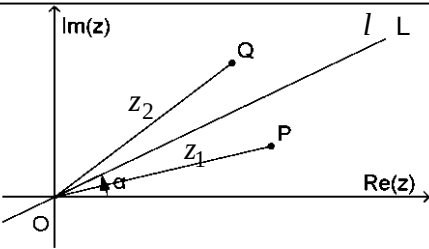
	When $x = 2$, $(2)^2 + 1 \equiv \frac{2}{3} \left((2)^2 + (2) + 1 \right) + \left(B(2) - \frac{1}{3} \right) ((2) - 1)$ $5 = \frac{14}{3} + 2B - \frac{1}{3}$ $5 = \frac{13}{3} + 2B$ $2B = \frac{2}{3}$ $B = \frac{1}{3}$ $\therefore \frac{x^2 + 1}{(x-1)(x^2 + x + 1)} \equiv \frac{\frac{2}{3}}{x-1} + \frac{\frac{1}{3}x - \frac{1}{3}}{x^2 + x + 1} \quad \textcircled{1} \quad \textcircled{1}$	
12.	(b)(ii) $\int \frac{x^2 + 1}{(x-1)(x^2 + x + 1)} dx$ $= \int \frac{\frac{2}{3}}{x-1} + \frac{\frac{1}{3}x - \frac{1}{3}}{x^2 + x + 1} dx$ $= \int \frac{\frac{2}{3}}{x-1} dx + \int \frac{\frac{1}{3}x - \frac{1}{3}}{x^2 + x + 1} dx$ $= \frac{2}{3} \int \frac{1}{x-1} dx + \frac{1}{3} \int \frac{x-1}{x^2 + x + 1} dx$ ① $= \frac{2}{3} \ln x-1 + \frac{1}{2} \left(\frac{1}{3} \right) \int \frac{2x-2}{x^2 + x + 1} dx$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \int \frac{2x+1-2-1}{x^2 + x + 1} dx$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \int \frac{2x+1}{x^2 + x + 1} - \frac{3}{x^2 + x + 1} dx$ ① $= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2 + x + 1) - \frac{1}{2} \int \frac{1}{x^2 + x + 1} dx$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2 + x + 1) - \frac{1}{2} \int \frac{1}{x^2 + x + \left(\frac{1}{2}\right)^2 + 1 - \frac{1}{4}} dx$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2 + x + 1) - \frac{1}{2} \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx$	

	$= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2+x+1) - \frac{1}{2} \left(\frac{2}{\sqrt{3}} \right) \int \frac{1 \times \frac{\sqrt{3}}{2}}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2+x+1) - \frac{1}{\sqrt{3}} \tan^{-1} \frac{\left(x + \frac{1}{2}\right)}{\frac{\sqrt{3}}{2}} + C$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2+x+1) - \frac{1}{\sqrt{3}} \tan^{-1} \frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right) + C$ $= \frac{2}{3} \ln x-1 + \frac{1}{6} \ln(x^2+x+1) - \frac{1}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} + C$	
12.	(c) Assume that there exists $p, q \in N$ such that $\log_2 5 = \frac{p}{q}$ and the highest common factor of p and q is 1. ① $5 = 2^{\frac{p}{q}}$ $(5)^q = \left(2^{\frac{p}{q}}\right)^q$ $\therefore 5^q = 2^p$ But LHS = 5^q $= 5 \times 5 \times 5 \times \dots$ i.e. 5 is a factor of 5^q but not 2^p and RHS = 2^p $= 2 \times 2 \times 2 \times \dots$ i.e. 2 is a factor of 2^p but not 5^q ① Then no p and q such that $p, q \in N$ satisfies the equation $5^q = 2^p$ and this equation must be a contradiction. So $\log_2 5$ is an irrational number. ①	

13.	(a)(i) $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \lambda + 1 \\ \lambda \\ 2\lambda + 3 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$	well done
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13.	(a)(ii) Equation of sphere $\left \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \right = \sqrt{29}$ $\left \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \right = \sqrt{29}$ $\left \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right = \sqrt{29}$ $\therefore (\lambda - 1)^2 + (\lambda + 1)^2 + (2\lambda + 3)^2 = 29$ $\therefore 6\lambda^2 + 12\lambda - 18 = 0$ $\therefore \lambda^2 + 2\lambda - 3 = 0$ $\therefore (\lambda + 3)(\lambda - 1) = 0$ $\therefore \lambda = -3, 1$ $\lambda = -3: P = (1 - 3, 0 - 3, 3 - 6) = (-2, -3, -3)$ $\lambda = 1: Q = (1 + 1, 0 + 1, 3 + 2) = (2, 1, 5)$	Mostly well done ① ① ①
13.	(a)(iii) $PQ = \sqrt{(-2 - 2)^2 + (-3 - 1)^2 + (-3 - 5)^2}$ $= \sqrt{16 + 16 + 64}$ $= \sqrt{96}$ $= 2\sqrt{24}$ $\neq 2\sqrt{29}$ Where $\sqrt{29}$ is the radius $\therefore PQ$ is not the diameter of the sphere.	Well done ①

13.	(b)(i) $x = 6 \cos\left(2t + \frac{\pi}{4}\right) + \cos(2t)$ $\dot{x} = -12 \sin\left(2t + \frac{\pi}{4}\right) - 2 \sin(2t)$ $\ddot{x} = -24 \cos\left(2t + \frac{\pi}{4}\right) - 4 \cos(2t)$ $= -4 \left[6 \cos\left(2t + \frac{\pi}{4}\right) + \cos(2t) \right]$ $\ddot{x} = -4x$ $\ddot{x} = -n^2 x$ Which is SHM about $c = 0$ and $n = 2 \therefore$ period $= \frac{2\pi}{2} = \pi$	Mostly well done ① ① ①
13.	(b)(ii) $x = 6 \cos\left(2t + \frac{\pi}{4}\right) + \cos(2t)$ $= 6 \left[\cos(2t) \cos \frac{\pi}{4} - \sin(2t) \sin \frac{\pi}{4} \right] + \cos(2t)$ $= \frac{6\sqrt{2}}{2} [\cos(2t) - \sin(2t)] + \cos(2t)$ $= (3\sqrt{2} + 1) \cos(2t) - 3\sqrt{2} \sin(2t)$ $= R \cos(2t + \alpha) \text{ where } R = \sqrt{(3\sqrt{2} + 1)^2 + (3\sqrt{2})^2}$ $= \sqrt{6\sqrt{2} + 37}$ $= 6.744$ $\therefore$ amplitude $= 6.7$ (1d.p)	Quite a few students did not realise that the amplitude could be found by simply ① finding the coefficient of the distance equation by using the transformation/auxiliary angle formula. This was a lot easier than finding when the velocity ① was zero and substituting back t into x. ①
13.	(c) $x = \sin \theta$ $dx = \cos \theta$ $\therefore \int \frac{dx}{(1-x^2)^3} = \int \frac{\cos \theta \cdot d\theta}{\sqrt{(1-\sin^2 \theta)^3}}$ $= \int \frac{\cos \theta \cdot d\theta}{(\cos^2 \theta)^{\frac{3}{2}}}$ let $= \int \frac{d\theta}{\cos^2 \theta}$ $= \int \sec^2 \theta \cdot d\theta$ $= \tan \theta + C$ $= \frac{x}{\sqrt{1-x^2}} + C$ 	Very well done ① ① ① ①

14.	(a)(i)  Let the line l passes through the point L such that $\angle LOP = \angle LOQ = \alpha - \arg(z_1)$ ① for any similar statement $\arg(z_2) = \angle QOL + \alpha$ statement $= (\alpha - \arg(z_1)) + \alpha$ $= 2\alpha - \arg(z_1)$ $\therefore \arg(z_1) + \arg(z_2) = 2\alpha$ (as required) ① including alternate methods	Mostly did well!
14.	(a)(ii) $z_1 z_2 = z_1 z_2$ and since $z_1 = z_2$ $\therefore z_1 z_2 = z_1 ^2$ $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$ $\arg(z_1 z_2) = 2\alpha$ [from part (i)] $\therefore z_1 z_2 = z_1 ^2 (\cos 2\alpha + i \sin 2\alpha)$. (as required) ①	
14.	(a)(iii) When $\alpha = \frac{\pi}{4}$, $z_1 z_2 = z_1 ^2 \left(\cos 2\left(\frac{\pi}{4}\right) + i \sin 2\left(\frac{\pi}{4}\right) \right)$ ① $= z_1 ^2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$ $z_1 z_2 = z_1 ^2 i$. $z_1 \neq 0$ otherwise $\arg(z_1)$ is undefined. (So the locus of R is dependent on z_1). Hence the locus of R is the positive y – axis, excluding the origin, of length $z_1 ^2$ ①	Many have problem with the locus of R . Few students answered partially correct not taking into account that $ z_1 ^2$ will only give positive values and forget that O should be excluded.
14.	(b)(i) $a^2 + b^2 > 2ab$ — ① $b^2 + c^2 > 2bc$ — ② $a^2 + c^2 > 2ac$ — ③ ① + ② + ③ $2a^2 + 2b^2 + 2c^2 > 2ab + 2bc + 2ac$ $2(a^2 + b^2 + c^2) > 2(ab + bc + ac)$ $\therefore a^2 + b^2 + c^2 > ab + bc + ac$ (as required) ①	All did well.

14.	(b)(ii) Since $a+b+c=6$ $(a+b+c)^2 = (6)^2$ $(a+b+c)^2 = 36$ $[(a+b)+c]^2 = 36$ $(a+b)^2 + 2(a+b)c + c^2 = 36$ $a^2 + 2ab + b^2 + 2ac + 2bc + c^2 = 36$ $(a^2 + b^2 + c^2) + 2ab + 2ac + 2bc = 36 \quad \text{--- ④} \quad \textcircled{1}$ And from (i) $a^2 + b^2 + c^2 > ab + bc + ac$ $a^2 + b^2 + c^2 + 2(ab + bc + ac) > ab + bc + ac + 2(ab + bc + ac)$ $a^2 + b^2 + c^2 + 2(ab + bc + ac) > 3(ab + bc + ac) \quad \text{--- ⑤}$ Sub ④ into ⑤ $36 > 3(ab + bc + ac)$ $\therefore ab + bc + ac < 12 \quad \textcircled{1}$	Mostly did well.
14.	(c) $\int \frac{1}{3 - \cos x - 2 \sin x} dx$ Let $t = \tan \frac{x}{2}$ $\frac{x}{2} = \tan^{-1} t$ $x = 2 \tan^{-1} t$ $\frac{dx}{dt} = \frac{2}{1+t^2}$ $dx = \frac{2}{1+t^2} dt$ And $\cos x = \frac{1-t^2}{1+t^2}$ $\sin x = \frac{2t}{1+t^2}$ $= \int \frac{1}{3 - \frac{1-t^2}{1+t^2} - 2\left(\frac{2t}{1+t^2}\right)} \left(\frac{2}{1+t^2}\right) dt \quad \textcircled{1}$ $= \int \frac{1}{\frac{3(1+t^2) - (1-t^2) - 2(2t)}{1+t^2}} \left(\frac{2}{1+t^2}\right) dt$ $= \int \frac{1+t^2}{3(1+t^2) - (1-t^2) - 2(2t)} \left(\frac{2}{1+t^2}\right) dt$ $= \int \frac{1+t^2}{3+3t^2-1+t^2-4t} \left(\frac{2}{1+t^2}\right) dt$ $= \int \frac{2}{4t^2 - 4t + 2} dt \quad \textcircled{1}$	Mostly did well. Some errors were: 1) answer in term of t and not x. 2) forgotten the existence of the first term 3 when working out the equivalent fraction with denominator $1+t^2$. 3) was stuck with working out $\int \frac{2}{4t^2 - 4t + 2} dt$

	$= \int \frac{2}{(4t^2 - 4t + 1) + 2 - 1} dt$ $= \int \frac{2}{(2t-1)^2 + 1} dt \quad \text{i.e. } f(x) = 2x - 1 \text{ and } a = 1$ $f'(x) = 2$ $= \tan^{-1}(2t - 1) + C$ $= \tan^{-1}\left(2 \tan \frac{x}{2} - 1\right) + C$	①
14.	(d) $v = \sqrt{16 - 3x^2}$ $\frac{dx}{dt} = \sqrt{16 - 3x^2}$ $\frac{1}{\sqrt{16 - 3x^2}} dx = dt$ $\int_2^x \frac{1}{\sqrt{16 - 3x^2}} dx = \int_0^t dt$ $\int_2^x \frac{1}{\sqrt{(4)^2 - (\sqrt{3}x)^2}} dx = [t]_0^t$ $\frac{1}{\sqrt{3}} \int_2^x \frac{\sqrt{3}}{\sqrt{(4)^2 - (\sqrt{3}x)^2}} dx = t - 0$ $\frac{1}{\sqrt{3}} \left[\sin^{-1} \frac{\sqrt{3}x}{4} \right]_2^x = t$ $\sin^{-1} \frac{\sqrt{3}x}{4} - \sin^{-1} \frac{\sqrt{3}(2)}{4} = \sqrt{3}t$ $\sin^{-1} \frac{\sqrt{3}x}{4} - \sin^{-1} \frac{\sqrt{3}}{2} = \sqrt{3}t$ $\sin^{-1} \frac{\sqrt{3}x}{4} - \frac{\pi}{3} = \sqrt{3}t$ $\sin^{-1} \frac{\sqrt{3}x}{4} = \sqrt{3}t + \frac{\pi}{3}$ $\frac{\sqrt{3}x}{4} = \sin\left(\sqrt{3}t + \frac{\pi}{3}\right)$ $\therefore x = \frac{4}{\sqrt{3}} \sin\left(\sqrt{3}t + \frac{\pi}{3}\right)$	Mostly did well especially using the boundaries. ① ① ①
14.	(e) $\underline{r}(t) = \begin{bmatrix} 10 \cos t \\ 10 \sin t \\ 15 - t \end{bmatrix}$	Mostly did well and a handful of students found the initial velocity vector but forgot to calculate the speed.

	$\therefore \underline{v}(t) = \begin{bmatrix} -10 \sin t \\ 10 \cos t \\ -1 \end{bmatrix} \quad \textcircled{1}$ $\underline{v}(0) = \begin{bmatrix} -10 \sin(0) \\ 10 \cos(0) \\ -1 \end{bmatrix}$ $\underline{v}(0) = \begin{bmatrix} 0 \\ 10 \\ -1 \end{bmatrix}$ $ \underline{v}(0) = \sqrt{0^2 + (10)^2 + (-1)^2}$ $ \underline{v}(0) = \sqrt{101} \text{ ms}^{-1} \quad \textcircled{1}$	
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15.	(a)(i) $z^5 + 1 = 0$ $\therefore z^5 = -1 \Rightarrow e^{i(\pi + 2\pi n)} \text{ where } n \in \mathbb{Z}$ So $z = e^{i\left[\frac{(\pi + 2\pi n)}{5}\right]}$ where $n = 0, 1, -1, 2, -2$. $\Rightarrow z = e^{\frac{i\pi}{5}}, e^{\frac{-i\pi}{5}}, e^{\frac{3i\pi}{5}}, e^{\frac{-3i\pi}{5}}, -1$	Mostly well done ① ①
15.	(a)(ii) $z^5 + 1 = 0$ $(z+1)(z^4 - z^3 + z^2 - z + 1) = 0$ now $z \neq -1$ $\therefore z^4 - z^3 + z^2 - z + 1 = 0$ $\therefore \frac{z^4 - z^3 + z^2 - z + 1}{z^2} = 0$ $\therefore z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$ $\therefore z^2 + 2 + \frac{1}{z^2} - \left(z + \frac{1}{z}\right) - 1 = 0$ $\therefore \left(z + \frac{1}{z}\right)^2 - \left(z + \frac{1}{z}\right) - 1 = 0$ sub $u = z + \frac{1}{z}$ $\therefore u^2 - u - 1 = 0 \text{ as req.}$	Mostly well done ① ①

15.	(a)(iii) $u^2 - u - 1 = 0$ $\therefore u = \frac{1 \pm \sqrt{5}}{2}$ now $u = z + \frac{1}{z}$ $= z + \bar{z}$ $= 2 \operatorname{Re}(z)$ $= 2 \cos \theta$ Hence $\cos \theta = \frac{u}{2} = \frac{1 \pm \sqrt{5}}{4}$ Now $\cos \frac{\pi}{5} > \cos \frac{3\pi}{5}$ (1st quad pos > 2nd quad neg) $\therefore \cos \frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}$	Mostly well done Some students missed the correct quadrant and did ① the + instead of the - ① ①
15.	(b)(i) If a complex number $a + ib$ where a, b are real, is a root of a polynomial equation with real coefficients, then its complex conjugate is also a root of this equation. Since a, b, c, d are integer coefficients as stated, then the conjugate of $0 + ki$, being $0 - ki$ is also a root.	well done ①
15.	(b)(ii) Since $x = ki$ is a root of $x^4 + ax^3 + bx^2 + cx + d = 0$ Then $(ki)^4 + a(ki)^3 + b(ki)^2 + c(ki) + d = 0$ i.e. $k^4 - ak^3i - bk^2 + cki + d = 0$ i.e. $(k^4 - bk^2 + d) + i(ck - ak^3) = 0 + 0i$ $\therefore ck - ak^3 = 0 \Rightarrow k(c - ak^2) = 0$ equating imag. parts $\therefore c = ak^2$ (as $k \neq 0$) as required.	Mostly well done. Some used a different method ① which was fine but led to problems as not knowing how to proceed correctly in parts (iii) and (iv) ①
	(b)(iii) From (ii) $k^4 - bk^2 + d = 0$ (equating real) and since $c = ak^2 \Rightarrow k^2 = \frac{c}{a}$ Then $\left(\frac{c}{a}\right)^2 - b\left(\frac{c}{a}\right) + d = 0$ $c^2 - abc + a^2d = 0$ $\therefore c^2 + a^2d = abc$ as req.	Mostly well done if used this method' ① ①
	(b)(iv) If $b = 0$, then from (iii) $c^2 + a^2d = 0$ then $d = -\left(\frac{c}{a}\right)^2$ and since d is integral, then c must be divisible by a, i.e. $c = ap$ where p is integral.	Mostly well done if used this method. There is no guarantee that roots are integral in saying

Then $d = -\left(\frac{ap}{a}\right)^2 = -p^2$ Also, the given equation $x^4 + ax^3 + bx^2 + cx + d = 0$ becomes $16 + 8a + 2c + d = 0$ since 2 is a root. From $16 + 8a + 2c + d = 0 \Rightarrow d = -2(8 + 4a + c)$ hence d is even as it divisible by 2 and 8, 4, a, c are integral. Since d is even and $d = -p^2$, it follows p must be even and hence $p = 2q \therefore d = -4q^2$ Now from $16 + 8a + 2c + d = 0$ we have $16 + 8a + 2c - 4q^2 = 0$ That is $c = -2(q^2 - 4 - 2a)$ which is even as required as q, a are integral.	① it is divisible by 2 which quite a few stated. i.e α and β etc ① ①
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16. (a)(i) $T_n = 2 \left[\frac{1 + (-3)^{-n}}{1 - (-3)^{-n}} \right]$ When $n = 1$, $T_1 = 2 \left[\frac{1 + (-3)^{-1}}{1 - (-3)^{-1}} \right]$ $= 2 \left[\frac{1 - \frac{1}{3}}{1 - \left(-\frac{1}{3}\right)} \right]$ $= 2 \left[\frac{\left(\frac{2}{3}\right)}{\left(\frac{4}{3}\right)} \right]$ $= 2 \left[\frac{1}{2} \right]$ $\therefore T_1 = 1$ which agrees with the recursive definition. Hence true for $n = 1$. Assume true for $n = 1$ to $n = k$, $T_1 = 2 \left[\frac{1 + (-3)^{-1}}{1 - (-3)^{-1}} \right]$ $T_2 = 2 \left[\frac{1 + (-3)^{-2}}{1 - (-3)^{-2}} \right]$	Mostly did well and a handful of students did not keep to the format for recursive method of induction. A few students had problem with working with base -3. ①
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$$T_3 = 2 \left\lfloor \frac{1 + (-3)^{-3}}{1 - (-3)^{-3}} \right\rfloor$$

$$\vdots \quad \quad \vdots$$

$$T_{k-1} = 2 \left\lfloor \frac{1 + (-3)^{-(k-1)}}{1 - (-3)^{-(k-1)}} \right\rfloor$$

$$T_k = 2 \left\lfloor \frac{1 + (-3)^{-k}}{1 - (-3)^{-k}} \right\rfloor$$

①

$$\text{Required to prove } T_{k+1} = 2 \left\lfloor \frac{1 + (-3)^{-(k+1)}}{1 - (-3)^{-(k+1)}} \right\rfloor$$

Proof: LHS = T_{k+1}

$$= \frac{4 + T_k}{1 + T_k} \quad (\text{from recursive definition})$$

$$= \frac{4 + 2 \left\lfloor \frac{1 + (-3)^{-k}}{1 - (-3)^{-k}} \right\rfloor}{1 + 2 \left\lfloor \frac{1 + (-3)^{-k}}{1 - (-3)^{-k}} \right\rfloor} \quad (\text{from assumption}) \quad \text{①}$$

$$= \frac{4 \left\lfloor 1 - (-3)^{-k} \right\rfloor + 2 \left\lfloor 1 + (-3)^{-k} \right\rfloor}{1 \left\lfloor 1 - (-3)^{-k} \right\rfloor + 2 \left\lfloor 1 + (-3)^{-k} \right\rfloor}$$

$$= \frac{4 \cdot 3^k - 4(-1)^k + 2 \cdot 3^k + 2(-1)^k}{3^k - (-1)^k + 2 \cdot 3^k + 2(-1)^k}$$

$$= \frac{6 \cdot 3^k - 2(-1)^k}{3 \cdot 3^k + (-1)^k}$$

$$= \frac{2 \left\lfloor 3 \cdot 3^k - (-1)^k \right\rfloor}{3 \cdot 3^k + (-1)^k}$$

$$= \frac{2 \left\lfloor \frac{3 \cdot 3^k - (-1)^k}{3 \cdot 3^k} \right\rfloor}{\frac{3 \cdot 3^k + (-1)^k}{3 \cdot 3^k}}$$

$$= \frac{2 \left\lfloor \frac{3 \cdot 3^k}{3 \cdot 3^k} - \frac{(-1)^k}{3 \cdot 3^k} \right\rfloor}{\frac{3 \cdot 3^k}{3 \cdot 3^k} + \frac{(-1)^k}{3 \cdot 3^k}}$$

	$= \frac{2 \left[1 - \frac{(-1)^k}{3^{k+1}} \right]}{1 + \frac{(-1)^k}{3^{k+1}}}$ $= \frac{2 \left[1 - \frac{(-1)^k}{3^{k+1}} \times \frac{-1}{-1} \right]}{1 + \frac{(-1)^k}{3^{k+1}} \times \frac{-1}{-1}}$ $= \frac{2 \left[1 + \frac{(-1)^{k+1}}{3^{k+1}} \times \frac{1}{1} \right]}{1 - \frac{(-1)^{k+1}}{3^{k+1}} \times \frac{1}{1}}$ $= \frac{2 \left[1 + \left(-\frac{1}{3} \right)^{k+1} \right]}{\left[1 - \left(-\frac{1}{3} \right)^{k+1} \right]}$ $= \frac{2 \left[1 + (-3)^{-(k+1)} \right]}{\left[1 - (-3)^{-(k+1)} \right]} \quad (\text{closed definition})$ $\therefore \text{LHS} = \text{RHS}$ If true for $n = k$, hence proven true for $n = k + 1$. Since true for T_1, hence proven true for $T_{1+1} = T_2$, $T_{2+1} = T_3$, and so on. Hence true for all positive integers $n \geq 1$. ①	
16.	(a)(ii) $\lim_{n \rightarrow \infty} \left(-\frac{1}{3} \right)^n = 0$ $\therefore \lim_{n \rightarrow \infty} T_n = \lim_{n \rightarrow \infty} \frac{2 \left[1 + \left(-\frac{1}{3} \right)^n \right]}{\left[1 - \left(-\frac{1}{3} \right)^n \right]}$ $= \frac{2[1+0]}{[1-0]}$ $= 2$ ①	Mostly did well.
16.	(b)(i) $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ i.e. $I_{n+2} = \int_0^{\frac{\pi}{4}} \tan^{n+2} x \, dx$	Mostly did well. Some students did not know their trig. Identity well enough that they missed the fact that derivative of $\tan x$ is $\sec^2 x$ and

	$\begin{aligned} \therefore I_{n+2} &= \int_0^{\frac{\pi}{4}} \tan^{n+2} x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x \tan^2 x \, dx \quad \textcircled{1} \text{ or other method} \\ &= \int_0^{\frac{\pi}{4}} \tan^n x (\sec^2 x - 1) \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x \sec^2 x - \tan^n x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} \tan^n x \, dx \\ I_{n+2} &= \left[\frac{\tan^{n+1} x}{n+1} \right]_0^{\frac{\pi}{4}} - I_n \\ I_{n+2} + I_n &= \left[\frac{\tan^{n+1} \left(\frac{\pi}{4} \right)}{n+1} - \frac{\tan^{n+1}(0)}{n+1} \right] \\ I_{n+2} + I_n &= \left[\frac{(1)^{n+1}}{n+1} - (0) \right] \\ \therefore I_{n+2} + I_n &= \frac{1}{n+1} \quad (\text{as required}) \quad \textcircled{1} \end{aligned}$	that $\tan^2 x = \sec^2 x - 1$, instead they were getting very lost by using integration by parts. 2 students did not answer the required expression of $I_{n+2} + I_n = \frac{1}{n+1}$
16.	(b)(ii) $I_4 = \int_0^{\frac{\pi}{4}} \tan^4 x \, dx$ $I_4 = \frac{1}{(2)+1} - I_2$ $I_2 = \frac{1}{(0)+1} - I_0$ $I_0 = \int_0^{\frac{\pi}{4}} \tan^0 x \, dx$ $= \int_0^{\frac{\pi}{4}} 1 \, dx$ $= [x]_0^{\frac{\pi}{4}}$ $= \frac{\pi}{4}$ $\therefore I_2 = 1 - \frac{\pi}{4}$ $\therefore I_2 = \frac{4 - \pi}{4}$	A few students had problems with trig. identity $\tan^2 x + 1 = \sec^2 x$ A few students made the mistakes in identifying n value for I_4 as 4, rather than 2, hence had $\frac{1}{n+1}$ as $\frac{1}{(4)+1}$ instead of $\frac{1}{(2)+1}$. ①

	$\therefore I_4 = \frac{1}{3} - \frac{4-\pi}{4} \quad \text{or} \quad \frac{1}{3} - \left(1 - \frac{\pi}{4}\right)$ $= \frac{4-3(4-\pi)}{12} \quad \text{or} \quad \frac{1}{3} - 1 + \frac{\pi}{4}$ $\therefore I_4 = \frac{3\pi-8}{12} \quad \text{or} \quad \frac{\pi}{4} - \frac{2}{3} \quad \textcircled{1}$	
16.	(b)(iii) $J_n = (-1)^n I_{2n}$ $J_{n-1} = (-1)^{n-1} I_{2(n-1)}$ $J_n - J_{n-1} = (-1)^n I_{2n} - (-1)^{n-1} I_{2(n-1)} \quad \text{for } n \geq 1$ $= (-1)^{n-1} [(-1)I_{2n} - I_{2(n-1)}]$ $= (-1)^{n-1} (-1) [I_{2n} + I_{2(n-1)}]$ $= (-1)^n [I_{2n} + I_{2(n-1)}] \quad \text{and from part (i)}$ $I_{n+2} + I_n = \frac{1}{n+1}$ $I_{(2n)+2} + I_{(2n)} = \frac{1}{(2n)+1}$ $\therefore I_{(2n)} + I_{(2n)-2} = \frac{1}{(2n-2)+1}$ $\therefore J_n - J_{n-1} = (-1)^n \left(\frac{1}{2n-1} \right) \quad \text{for } n \geq 1$	“Show” question: a few students skipped steps.
16.	(b)(iv) From part (iii) $J_n - J_{n-1} = (-1)^n \left(\frac{1}{2n-1} \right)$ $\therefore J_m = (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3})$ $+ \dots + (J_1 - J_0) + J_0$ $= J_0 + (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3})$ $+ \dots + (J_1 - J_0)$ $= J_0 + (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3})$ $= J_0 + (-1)^1 \left(\frac{1}{2(1)-1} \right) + (-1)^2 \left(\frac{1}{2(2)-1} \right)$ $+ (-1)^3 \left(\frac{1}{2(3)-1} \right) + \dots + (-1)^m \left(\frac{1}{2m-1} \right) \quad \textcircled{1}$ $= (-1)^0 I_{2(0)} + \sum_{n=1}^m (-1)^n \left(\frac{1}{2n-1} \right) \quad \textcircled{1}$ $= I_0 + \sum_{n=1}^m \frac{(-1)^n}{2n-1}$	Mostly did well. Easy question to show since the series for J_m. Students need to show the substitution of $J_n - J_{n-1} = (-1)^n \left(\frac{1}{2n-1} \right)$ for the sigma notation, and either show how $J_0 = \frac{\pi}{4}$.

	$\therefore J_m = \frac{\pi}{4} + \sum_{n=1}^m \frac{(-1)^n}{2n-1} \quad (\text{as required})$	
16.	(b)(v) $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ Let $u = \tan x$ $\frac{du}{dx} = \sec^2 x$ $\frac{du}{dx} = \tan^2 x + 1$ $\frac{dx}{du} = \frac{1}{\tan^2 x + 1}$ $dx = \frac{1}{u^2 + 1} du$ When $x = \frac{\pi}{4}$, $u = 1$ $x = 0$, $u = 0$ $\therefore I_n = \int_0^1 u^n \frac{du}{1+u^2}$ $\therefore I_n = \int_0^1 \frac{u^n}{1+u^2} du \quad (\text{as required}) \quad \textcircled{1}$	Mostly did well.
16.	(b)(vi) From (i) $I_n + I_{n+2} = \frac{1}{n+1}$. Since $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx > 0$, and $I_{n+2} = \int_0^{\frac{\pi}{4}} \tan^{n+2} x \, dx > 0$ Then $I_n < \frac{1}{n+1}$ and $I_{n+2} < \frac{1}{n+1}$ Hence $0 < I_n < \frac{1}{n+1}$. ① As $\lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$ i.e. $\lim_{n \rightarrow \infty} I_n = 0$ and $\lim_{n \rightarrow \infty} I_{2n} = 0$ and $\lim_{n \rightarrow \infty} J_n = \lim_{n \rightarrow \infty} (-1)^n I_{2n}$ ① $\therefore \lim_{n \rightarrow \infty} J_n = 0 \quad \left(\text{and } \frac{\pi}{4} + \sum_{n=1}^m \frac{(-1)^n}{2n-1} = 0 \right)$ $\frac{\pi}{4} = - \sum_{n=1}^m \frac{(-1)^n}{2n-1}$	Some students have approached from different perspective and did well. Students that did not give a comprehensive approach for either or both deductions were not awarded the marks.

	$\frac{\pi}{4} = \sum_{n=1}^m \frac{(-1)^{n+1}}{2n-1}$	
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